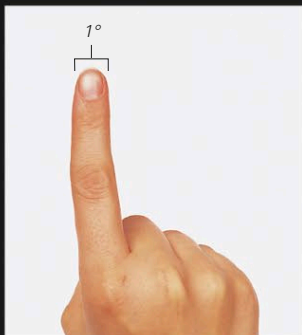


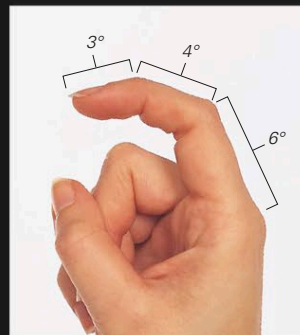
MEASUREMENTS ON THE SKY

Distances between objects in the sky are often expressed as degrees of angle. All the way around the horizon measures 360° , while the angle from horizon to zenith (the point directly overhead) is 90° . The Sun and Moon both have an angular diameter of 0.5° , while an outstretched hand can be used to estimate other distances. When studying star charts, bear in mind that 1 hour of right ascension (RA) along the celestial equator is equivalent to 15° of declination (see p.63), but right ascension circles get tighter toward the celestial poles, so at 60°N , an hour's difference in RA is equivalent to only 7.5° of declination.



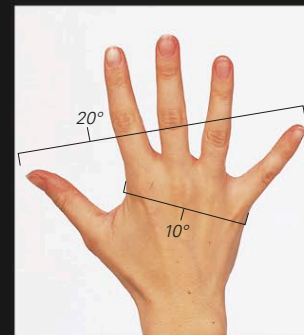
FINGER WIDTH

Held out at arm's length, a typical adult index finger blocks out roughly 1 degree of the sky—enough to cover the Moon twice over.



FINGER JOINTS

The finger joints provide measures for distances of a few degrees. A side-on fingertip is about 3° wide, the second joint 4° , and the third 6° .



HAND SPANS

The hand (not including the thumb) is about 10° across at arm's length, while a stretched hand-span covers 20° of sky.

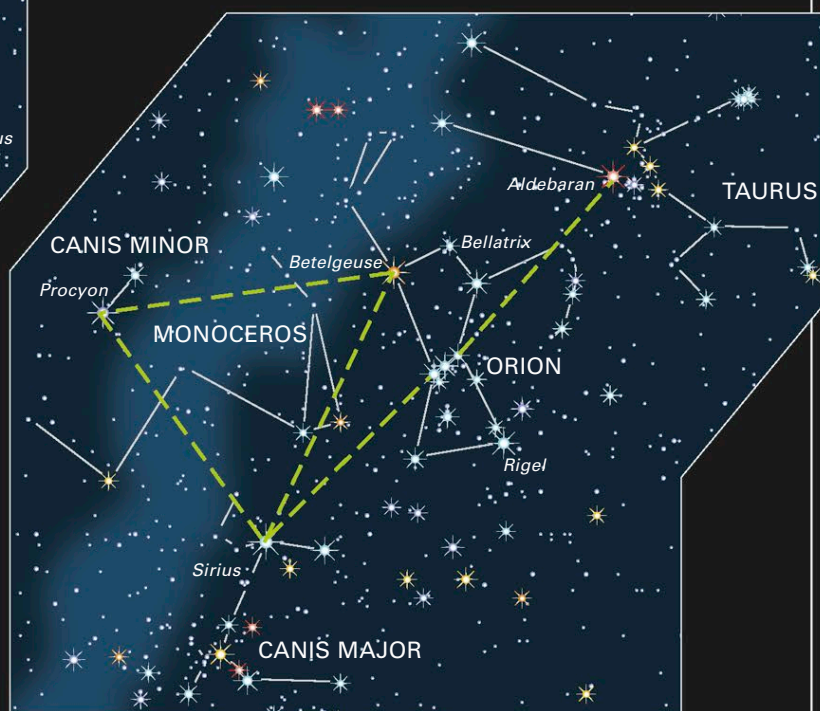


STAR HOPS FROM THE BIG DIPPER

A line through Dubhe and Merak along one side of the Big Dipper points straight to Polaris in one direction and (allowing for the curvature of the sky) toward the bright star Regulus in Leo in the other direction. Following the curve of the Big Dipper's handle, meanwhile, leads to the bright red star Arcturus in Boötes and eventually to Spica in Virgo.

ORION'S BELT AND THE WINTER TRIANGLE

The distinctive line of three bright stars forming Orion's belt points in one direction toward the red giant Aldebaran in Taurus and in the other toward Sirius, the brightest star in the sky, in Canis Major. Sirius, Betelgeuse (on Orion's shoulder), and Procyon (in Canis Minor) make up the equilateral Winter Triangle.



STAR-HOPPING

The best way to learn the layout of the night sky is to first find a few bright stars and constellations, then work outward into more obscure areas. Two key regions are the Big Dipper (the brightest seven stars in the constellation Ursa Major, close to the north celestial pole) and the area around the brilliant constellation Orion, including the Winter Triangle (see p.436) on the celestial equator. By following lines between certain stars in these constellations, one can find other stars and begin to learn the sky's overall layout. The Big Dipper is a useful pointer, because two of its stars align with Polaris, the star that marks the north celestial pole. Because the sky seems to revolve around the celestial poles, Polaris is the one fixed point in the northern sky. (There is no bright south Pole Star.) Other useful keystones are the Summer Triangle (see p.466)—comprising the northern stars Vega, Deneb, and Altair—and the Southern Cross (see p.437) and False Cross (see p.443) in the far south.